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Department of **Electronics & Electrical Engineering / Mechatronics Engineering**

Time Series Forecasting with FB Prophet and SARIMAX on Electricity Consumption in Romania produced by Energy

Final Report

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ABSTRACT

This project focuses on forecasting electricity consumption in Romania using two advanced time series models: Facebook Prophet and SARIMAX. The analysis covers both renewable energy sources—such as solar, wind, hydroelectric, and biomass—and non-renewable sources, including coal, gas, oil, and nuclear energy. The aim is to accurately predict energy consumption patterns by capturing seasonal trends, external influences like weather conditions, and long-term energy demand. By leveraging historical energy production and consumption data, the project evaluates the strengths of each model in managing the complexities of energy forecasting. Facebook Prophet is particularly suited for datasets with irregular seasonal patterns and minimal parameter tuning, making it a strong candidate for renewable energy forecasting, where variability is common. In contrast, SARIMAX offers robust performance when dealing with more stable datasets and exogenous variables, such as economic factors or fuel prices. The results of the study provide valuable insights into optimizing energy grid management, enhancing energy storage, and developing demand-response strategies. Moreover, accurate forecasts can assist policymakers in designing data-driven energy policies, reducing dependency on non-renewable sources, and improving the overall sustainability of Romania's energy infrastructure. This project emphasizes the critical role of predictive modeling in promoting energy efficiency, ensuring grid stability, and advancing the integration of both renewable and non-renewable energy sources.

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Chapter 1

Introduction

1.1 Background

Since the 1990s, climate change driven by the excessive buildup of greenhouse gases in Earth's atmosphere has become a critical global issue, drawing widespread attention and concern. Over the decades, greenhouse gas emissions have persistently risen, exacerbating the threat to the planet’s climate systems. The unchecked accumulation of these emissions has the potential to trigger severe and irreversible climate changes, impacting ecosystems, weather patterns, and human livelihoods across the globe. Consequently, reducing the concentration of greenhouse gas emissions is not just an environmental priority but a vital strategy in the global fight against climate change. Effective mitigation efforts, including transitioning to renewable energy sources, enhancing energy efficiency, and promoting sustainable practices, are essential to curbing emissions and safeguarding the planet's future. (Hadi Sasana, Panji Kusuma Prasetyanto, Diah Lufti Wijayanti, Ari Nurul Fatimah, 2023)

Production activities are widely recognized as a key contributor to greenhouse gas emissions. In today's world, energy consumption is essential for powering production processes, as it significantly enhances a nation's economic productivity. As a result, middle-income countries have been steadily increasing their energy consumption to stimulate and sustain economic growth, often pushing consumption levels to their peak. However, this heightened reliance on energy comes with a considerable downside, as it tends to degrade environmental quality. The increased demand for energy, particularly from non-renewable sources, leads to higher greenhouse gas emissions, exacerbating the challenges of climate change. Balancing the need for economic development with environmental sustainability has thus become a critical issue, requiring the adoption of cleaner energy sources and more efficient technologies to minimize the environmental impact while supporting continued growth. (Al-Mulali, U., Che Sab, C.N.B., 2018)

It is projected that the overall energy demand will increase more than twofold by 2050, and by the end of the century, it is estimated to triple due to rapid global industrialization. The environmental harm, depletion of fossil fuel reserves, health risks, and global climate change threats associated with excessive use of fossil fuels are recognized. To address these issues, improving current energy infrastructures alone is insufficient. Securing a sustainable energy supply for the future is one of the most challenging tasks society faces. Renewable energy

systems can provide solutions to modern energy challenges, such as increasing energy supply reliability and mitigating regional energy deficits. (Vishesh Mantri, NamoNarayan Sharma, Ramkumar Jayaraman, 2022)

Renewable energy is a vital and sustainable resource that offers an infinite supply and is markedly different from traditional energy sources due to its diversity, widespread availability, and minimal carbon dioxide emissions. Solar energy is harnessed from sunlight through silicon cells that convert it into electricity, with energy often stored in batteries for later use. Wind energy is generated by capturing the kinetic energy of moving air with wind turbines, and its efficiency improves with higher wind speeds. Geothermal energy comes from the Earth's internal heat, accessed by drilling wells into geothermal reservoirs to produce either direct heat or electricity. Biomass energy is created from organic materials like plant and animal waste, converted through processes such as combustion or fermentation into usable power. Each type of renewable energy utilizes distinct natural processes to generate power, contributing to a diverse and eco-friendly energy mix. (Ahmed M.A. Nahhas, Akaehomen O.Akii Ibhadode, 2023)

Energy prediction is crucial in the face of growing energy demands because it enables more efficient management and allocation of resources, ensuring that the energy supply can meet future needs without causing excessive environmental harm. Accurate energy forecasting helps governments and industries plan for energy production, optimize grid operations, and integrate renewable energy sources effectively. As energy demands increase, particularly in emerging economies striving for economic growth, the ability to predict energy requirements becomes vital to avoid shortages, reduce costs, and support sustainable development. Additionally, by anticipating future energy needs, stakeholders can invest in infrastructure and technology that aligns with long-term sustainability goals, mitigating the impact of energy production on climate change. (Megha Sharma, Namita Mittal, Anukram Mishra, Arun Gupta, 2023)

The utilization of machine learning techniques has significantly increased in many industries that have encountered data-driven challenges in recent years. These techniques are designed to identify relationships between input and output data, regardless of mathematical problem structures. Regression, a well-known supervised learning method, is a popular machine learning technique that predicts a target value based on independent variables. It is frequently used in forecasting to comprehend the associations between various variables. Regression models may differ depending on the type of relationship between

dependent and independent variables, as well as the number of independent variables evaluated. The primary aim of this analysis is to discover and establish the connections between input and output variables. (Mohit Kumar Shakya, Vijay N.Pande, Rohan S.Kulkarni, Suhas Kakade, 2023)

1.2 Motivation

Time series forecasting can significantly impact the electricity consumption sector by providing accurate predictions of energy usage from renewable sources. This technology helps optimize the integration of renewable energy into the grid, ensuring efficient energy distribution and reducing reliance on non-renewable sources. Accurate consumption forecasts can support participation in energy markets, guide investment decisions, and improve demand response management. In the long term, these forecasting solutions contribute to building a cleaner, more sustainable, and resilient energy system powered by renewables.

1.3 Problem Statement

Accurate forecasting and analysis of electricity consumption produced by various energy sources, such as solar, wind, nuclear, hydroelectric, biomass, coal, gas, and oil, present complex challenges due to the unique characteristics of each source. Solar and wind energy are intermittent and highly dependent on environmental conditions, requiring advanced machine learning models to predict consumption patterns and ensure grid stability. Hydroelectric and biomass energy production is influenced by seasonal and resource availability, necessitating time series forecasting to account for fluctuations in water levels and organic material. On the other hand, nuclear power provides a stable supply, but optimizing its consumption for efficiency is crucial. Fossil fuels, including coal, gas, and oil, contribute significantly to electricity generation but pose environmental concerns. Therefore, monitoring consumption trends and implementing strategies to reduce dependence on these sources while transitioning to cleaner energy remains essential for sustainable development.

1.4 Aims & Objectives

The objectives of time series forecasting regarding renewable energies are multifaceted and encompass the following:

1. **Enhancing Precision**: Refine and develop forecasting models, such as ARIMA and Facebook Prophet (FBP), to enhance the precision of predicting renewable energy generation levels and patterns over different time frames, ranging from short-term (hours to days) to long-term (weeks to months).
2. **Boosting Model Resilience**: Increase the resilience of forecasting models, particularly ARIMA and FBP, to account for the unpredictability and variability associated with renewable energy sources, including elements such as weather conditions, geographic variations, and atmospheric phenomena.
3. **Enabling Market Involvement**: Utilize accurate forecasts generated by ARIMA and FBP models of renewable energy output to enable energy producers to participate in energy markets and trading activities. These models empower producers to make informed decisions about energy trading, pricing, and revenue optimization.
4. **Promoting Sustainability and Durability**: Foster a sustainable and durable energy infrastructure by employing ARIMA and FBP to promote the adoption of renewable energy sources, decrease reliance on fossil fuels, mitigate greenhouse gas emissions, and advance towards a cleaner and more sustainable energy future.

1.5 Report Outline

This report has been thoughtfully organized into five distinct chapters, each focusing on a specific aspect of the research project. Chapter 1 introduces the background, motivation, and problem statement for forecasting electricity consumption from renewable energy sources, outlining the aims, objectives, and report structure. Chapter 2 reviews relevant literature, including conference proceedings and previous research, and provides a block diagram of the forecasting model. Chapter 3 discusses the dataset used for the project and details the machine learning models employed, such as ARIMA, SARIMAX, and FB Prophet, along with data preparation techniques including collection and preprocessing. It also covers model confirmation and performance metrics. Chapter 4 analyzes the performance of the models, focusing on train-test split, cross-validation method which is time series cross validation, and parameter tuning. Finally, Chapter 5 concludes the report with a summary of key findings, expected outcomes, and potential applications of the forecasting models.

Chapter 2

Literature Review

2.1 Review on Conference Proceedings

This literature review provides an overview of recent research and advancements in solar irradiation forecasting. By examining existing studies and findings, this review aims to offer insights into the current state of the field and identify areas for further investigation.

The rising demand for energy has become a major concern for the world, leading to the integration of solar energy into electric grids. Solar energy is a renewable source of energy that can effectively reduce the dependence on fossil fuels, which are non-renewable and harmful to the environment. In order to optimize the efficiency of photovoltaic (PV) energy systems, accurate forecasting is essential. Precise predictions of solar irradiation variations can help in better energy planning and distribution, leading to improved service quality and reduced costs. Solar irradiation forecasting involves the use of advanced technology and algorithms that take into account various factors such as the weather, time of day, and location of the PV system. By accurately predicting the amount of energy that can be generated by the PV system, it is possible to ensure that the energy is used efficiently and effectively. This can help in reducing the overall cost of energy and make it more accessible to a larger population. (Vishesh Mantri, NamoNarayan Sharma, Ramkumar Jayaraman, 2022)

Time series forecasting has a variety of prediction models, each with its advantages and limitations. Statistical models like ARIMA, Prophet, LR, and ES are popular and have shown good performance with univariate data. However, to use them effectively, domain knowledge and data distribution assumptions are needed, which limits their applicability to certain contexts. Meanwhile, machine learning models like neural networks (NNs), support vector regression (SVR), and random forests (RF) can learn nonlinear relationships between variables without pre-assumptions about the data, making them more flexible and widely applicable. Finally, deep learning models like long short-term memory (LSTM) and convolutional neural networks (CNN) are effective at capturing complex patterns in large datasets, making them well-suited for applications that require high accuracy and robustness. (Sarah Almaghrabi, Mashud Rana, Margaret Hamilton, Mohammad Saiedur Rahaman, n.d.)

PFM was introduced by Facebook in 2017 as a tool that models time series datasets. It can accommodate trends, seasonality, and holidays, and takes just a few seconds to fit the model. Various parameters can be optimized for improved forecasting. The model type can be

linear or logistic, where linear models normalize outliers and do not set maximum or minimum limits. Logistic models, on the other hand, are used for saturated forecasts, where the highest and lowest values are specified. Linear models are generally preferred for air pollution datasets since they account for the typical outliers observed in such trends. PFM's changepoints are crucial parameters. You can either use an automatic function to detect changepoints or specify the number of changepoints or scale of fit. A larger number of change points leads to a better fit, but the model may not be as effective at predicting future trends. When fewer changepoints are used, the model may be underfitted and too general, leading to an inaccurate graphical model. To determine the number of changepoints, PFM initially plots a large number and then uses L1 regularization to select the few changepoints that will be used in the model. This ensures that only the significant changepoints are selected and the number of changepoints is kept to a minimum. (Justin Shen, Davesh Valagolam, Serena McCalla, 2020)

(Asim Shakeel, Daotong Chong, Jinshi Wang, 2023) have investigated the application of the Facebook Prophet model in the context of renewable energy forecasting, showcasing its strengths in managing time series data with complex seasonal patterns and external influences. Their study demonstrates that Prophet, developed by Facebook, is particularly effective for renewable energy applications due to its capability to model both annual and weekly seasonality, as well as incorporate holiday effects and external regressors. The research highlights Prophet's flexibility and robustness, particularly in handling data with strong seasonal variations, such as those seen in renewable energy sources like solar and wind power. By leveraging an additive model approach, Prophet can decompose time series data into trend, seasonal, and holiday components, which enhances its ability to capture and forecast complex patterns. This characteristic is crucial for renewable energy forecasting where consumption and generation patterns are influenced by seasonal changes, weather conditions, and other external factors. They emphasize Prophet's ease of use and automated nature, which facilitates accurate forecasting even with minimal parameter tuning. The model's ability to generate uncertainty intervals further supports its application in decision-making processes, providing a measure of confidence in forecasts. This is particularly beneficial for energy managers and planners who need reliable predictions to optimize resource allocation and manage grid integration effectively. Their analysis confirms that Prophet's combination of automatic seasonality detection, adaptability to external variables, and ease of implementation makes it a valuable tool for renewable energy forecasting. The study underscores Prophet's role in improving forecast accuracy and supporting sustainable energy practices, contributing to better management of renewable energy resources and enhanced planning for energy systems.

(Ruixia Suo, Qi Wang, Yuanyuan Tan & Qiutong Han, 2024) have conducted an in-depth analysis of the ARIMA model in their study, focusing on its application for forecasting renewable energy consumption. Their research highlights ARIMA's foundational strengths in time series analysis, particularly its efficacy in modelling and predicting data with underlying trends and autoregressive structures. The study reveals that ARIMA remains a robust and widely used model for time series forecasting due to its ability to capture and forecast data patterns based on past values and trends. They emphasize ARIMA's effectiveness in handling non-stationary time series data by differentiating the data to achieve stationarity, which is crucial for accurate forecasting. This capability makes ARIMA particularly useful for renewable energy applications where data often exhibit complex temporal dependencies and trends. Suo, Wang, Tan, and Han also explore how ARIMA's autoregressive and moving average components enable it to model the dependencies between past observations and forecast future values. This aspect of ARIMA is particularly beneficial for predicting renewable energy consumption, which can be influenced by various temporal factors such as seasonal variations and historical consumption patterns. The model's capacity to provide forecasts based on these dependencies supports more reliable and actionable predictions for energy management and planning. They also discuss the practical implementation of ARIMA, including model selection and parameter tuning. Suo et al. highlight the importance of diagnostic checks and model validation to ensure the robustness of forecasts. They argue that while ARIMA is effective, its performance can be enhanced when combined with other techniques or adjusted for specific characteristics of renewable energy data.

(Elamin, Niematallah ; Fukushige, Mototsugu, 2018) have explored the application of SARIMAX model in forecasting renewable energy consumption, highlighting its effectiveness in managing both seasonal variations and external influences. Their study demonstrates how SARIMAX, an extension of the ARIMA framework, is particularly well-suited for modeling renewable energy data, which often exhibits pronounced seasonal patterns and is impacted by external factors such as weather conditions and economic policies. Elamin and Fukushige provide a comprehensive analysis of SARIMAX's capability to integrate seasonal components and external regressors, allowing for a nuanced understanding of energy consumption dynamics. The model's ability to account for multiple seasonal periods and external variables enhances forecasting accuracy, which is crucial for managing the variability and intermittency associated with renewable energy sources like wind and solar power. By accurately capturing cyclical behaviors and adapting to external changes, SARIMAX offers a more precise forecasting tool compared to simpler models. Their research underscores SARIMAX's practical advantages in real-world applications, including improved decision-making in energy

management and planning. The model's flexibility in handling complex, multi-seasonal data and adjusting for external factors facilitates better grid integration, resource optimization, and policy development. Elamin and Fukushige's findings affirm SARIMAX's role in advancing forecasting methods for renewable energy consumption, contributing to more reliable and actionable forecasts and supporting the broader goal of enhancing sustainable energy practices.

(Rahul Gupta, Anil Kumar Yadav, SK Jha, Pawan Kumar Pathak, 2022) presents a multivariate regression method for predicting the daily output power of grid-connected PV systems. It provides an overview of several techniques, such as support vector machines, ARIMA, LSTM, and deep convolutional neural networks, for forecasting PV power. Furthermore, it examines the use of FB Prophet and ARIMA models for predicting the output power of a 100kW PV plant. It also discusses how deep learning models, like convolutional neural networks and long short-term memory networks, can be utilized to predict solar irradiance. As renewable energy generation systems are variable in nature, accurate forecast measures are essential for smart grid infrastructure planning. The article focuses on LSTM networks for analysing sequential data. Additionally, it employs machine learning techniques, such as artificial neural networks, deep learning, support vector machines, and k-nearest neighbour algorithms, for forecasting global solar irradiance. The paper introduces a novel technique that employs the least absolute shrinkage and selection operator (LASSO) for accurate solar power generation forecasting. It also proposes an innovative approach involving modelling PV cell temperature and irradiance components. The article utilizes neural networks and machine learning algorithms for forecasting and nowcasting PV power generation, providing an extensive review across various time intervals. Finally, the paper suggests employing support vector machines as a predictive method for short-term forecasting of PV power output.

2.2 Block Diagram of the Forecasting Model

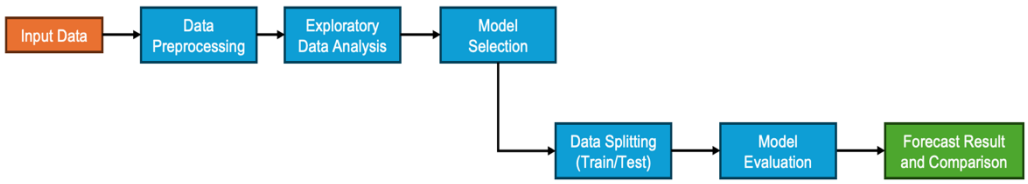

Figure 1 Block Diagram of the Forecasting Model

The time series forecasting process begins with the collection of relevant historical data, which is then subjected to data preprocessing steps such as cleaning, handling missing values, and normalization. Following this, exploratory data analysis (EDA) is conducted to identify patterns, trends, and seasonality within the data. Based on the insights from the EDA, appropriate forecasting models are selected. The dataset is subsequently split into training and testing subsets, allowing the model to be trained on a portion of the data and validated on unseen data to ensure robust performance. After training, the model's accuracy and reliability are assessed through model evaluation metrics. Finally, forecasts are generated and analyzed, with results compared to validate the model's effectiveness. This structured approach ensures that the forecasting process is both systematic and rigorous, leading to reliable predictive outcomes.

Chapter 3

Material and ML Methods for Data Prediction

3.1 Dataset Used for the Project

The dataset used in this project encompasses Romania's renewable energy production and consumption, offering valuable insights into the country's energy landscape. Romania experiences a continental climate, marked by distinct seasons that significantly influence renewable energy generation. Summers are typically hot, with temperatures often exceeding 30°C, particularly in the southern and eastern regions. These high temperatures are accompanied by long daylight hours, providing an ideal environment for solar energy production. However, the potential for heatwaves during summer can lead to variability in solar power output, as extremely high temperatures may reduce the efficiency of photovoltaic systems.

Winters in Romania are cold, with temperatures frequently dropping below freezing, especially in the mountainous regions and the north. Snowfall is common, and these cold conditions can impact the performance of solar panels by reducing sunlight exposure and potentially causing snow cover on the panels. On the other hand, the winter season is also characterized by strong winds, particularly in the southeastern region, making it a favourable period for wind energy generation. Romania's wind farms, located mainly along the Black Sea coast and in the Dobrogea region, benefit from these seasonal wind patterns, contributing significantly to the country's renewable energy output. Spring and autumn serve as transitional seasons, with moderate temperatures and variable weather conditions, including occasional rainfall and wind. These periods are crucial for understanding the seasonal shifts in energy production, as they provide a mix of solar and wind resources that can be harnessed for energy generation. (Meteo.ro, n.d.)

The analysis of Romania's renewable energy data, considering these diverse weather conditions, aims to develop robust forecasting models. These models will help predict future energy production and consumption patterns, considering the influence of seasonal variations and extreme weather events. The insights gained from this project will be instrumental in optimizing energy production, ensuring a reliable energy supply, and supporting Romania's efforts toward sustainable energy management and climate resilience.

3.2 Model Confirmation

For the time series forecasting of Romania's renewable energy consumption, two distinct models will be employed: Facebook Prophet and SARIMAX. Facebook Prophet will be used for its robustness in handling seasonal effects and incorporating holidays or special events that may impact energy consumption patterns. Meanwhile, the SARIMAX model will be leveraged for its capability to account for both seasonal and non-seasonal components in the data, as well as its flexibility in integrating exogenous variables. Utilizing both models aims to enhance forecasting accuracy and capture the intricate seasonal and trend patterns inherent in Romania's energy consumption data. This dual-model approach will facilitate a comprehensive analysis and provide a more reliable prediction framework for future energy needs.

FB Prophet	SARIMAX
Pros	
Handles missing data and outliers effectively	Flexible in modeling both seasonal and non-seasonal components
Captures seasonality and holidays well	Can include exogenous variables (predictors) for improved accuracy
Requires minimal parameter tuning	Effective for complex seasonal patterns and longer time series
User Friendly	
Cons	
Limited flexibility for complex seasonal patterns	Requires careful parameter tuning
May not perform well with highly irregular or sparse data	Computationally intensive for large datasets
Assumes a linear trend, which might not fit all datasets	Sensitive to data quality and may overfit without proper regularization

3.2.1 ARIMA

Time series data can be used to create a model that analyzes intrinsic trends and predicts future developments. The ARIMA model is made up of three components: autoregression (AR), integration (I), and moving average (MA). Autoregression (AR) models the connection between current and previous observations. Integration (I) transforms the time series into a stationary form, which is essential for building an econometric model. Moving average (MA) accounts for random fluctuations in the current observation. If the time series shows nonstationary behavior, as indicated by a unit root test, differencing must be used to transform it into a stationary form. This process guarantees that the ARIMA model can accurately capture and analyze the underlying data trends, making it a reliable tool for time series analysis. (Jiao Xie, Zhang Kanghuai, Yan Cai, Na Wang, Haitao Wang, Yang Zhao, Youjia Li, Xin Yang, Meixi Ji, Lijing Sun, and Jinqiu Wang, 2020)

The equation for ARIMA (p,d,q) model is defined as:

$$\hat{y}_t = \mu + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} - \theta_1 e_{t-1} - \cdots - \theta_q e_{t-q} + e_t$$

where:

$\hat{y}_t$ = forecasted value of time series at time t

μ = constant term (mean of the series)

$\phi_1, \phi_2, \ldots, \phi_p$= coefficients of the autoregressive (AR) terms

$\theta_1, \theta_2, \ldots, \theta_q$= coefficients of the moving average (MA) terms

e_t= white noise error term at time t

p = number of autoregressive terms (lags)

q = number of moving average terms (lags)

3.2.2 SARIMAX

SARIMAX (Seasonal Autoregressive Integrated Moving Average with exogenous factors) model is an advanced extension of the ARIMA model, tailored for time series data that exhibits both seasonal patterns and is influenced by external variables. It enhances the traditional ARIMA framework by incorporating seasonal components—such as seasonal autoregressive, differencing, and moving average terms—allowing it to model repeating cycles like monthly or yearly fluctuations effectively. Additionally, SARIMAX includes exogenous factors, which are external variables that may impact the time series, such as weather conditions, economic indicators, or promotional events. This integration of external influences enables SARIMAX to produce more accurate and contextually aware forecasts, especially in complex scenarios like energy demand forecasting, where consumption patterns are shaped by both seasonal trends and external factors like temperature. By capturing both the underlying seasonality and the effects of external variables, SARIMAX provides a flexible and powerful tool for generating precise forecasts across various domains. (Box, George E. P.; Jenkins, Gwilym M.; Reinsel, Gregory C., 2008)

$$Y_t = \mu + \phi(B)\Phi(B^s)\Delta^d\Delta_s^D Y_t + \theta(B)\Theta(B^s)\varepsilon_t + X_t\beta + \varepsilon_t$$

where:

Y_t = value of time series at time t

μ = constant term (optional)

$\phi(B)$= autoregressive (AR) polynomial of order p

$\Phi(B^s)$ = seasonal autoregressive polynomial of order P (seasonal period s)

Δ^d = differencing operator applied d times

Δ_s^D= seasonal differencing operator applied D times (seasonal period s)

$\theta(B)$ = moving average (MA) polynomial of order q

$\Theta(B^s)$ = seasonal moving average polynomial of order Q (seasonal period s)

X_t= vector of external regressors (exogenous variables) at time t

β = vector of coefficients for the exogenous variables

ε_t = error term or white noise at time t

3.2.3 FB Prophet

Facebook Prophet is an open-source forecasting tool developed by Facebook, designed to address the challenges of forecasting time series data that exhibits strong seasonal patterns, trends, and is often characterized by missing data or outliers. Prophet employs an additive model, where non-linear trends are combined with seasonal effects, allowing it to effectively capture daily, weekly, and yearly seasonality, as well as special events like holidays. One of the key strengths of Prophet is its ability to automatically handle missing data and outliers, making it robust in real-world applications where data quality may be inconsistent. The model is built with business forecasting in mind, which often involves irregular schedules, abrupt changes, and seasonal variations. It is highly flexible, allowing users to specify custom seasonality and account for known changepoints where the trend might shift. Prophet also provides uncertainty intervals around its forecasts, helping users gauge the confidence in the predictions, which is crucial for decision-making in uncertain environments.

Despite its sophisticated capabilities, Prophet is designed to be user-friendly, requiring minimal tuning or parameter adjustments. This makes it accessible to both data scientists and non-experts who need reliable forecasts without needing in-depth knowledge of time series analysis. Common applications of Prophet include forecasting sales, web traffic, and energy consumption, particularly in situations where data exhibits regular, cyclic patterns and where it's important to understand the impact of holidays or events on the forecast. Its versatility and ease of use have made Prophet a popular choice in industries where accurate and automated forecasting is essential.

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t$$

where:

y(t) = forecasted value at time t

g(t) = trend component, which models the non-periodic changes in the time series

s(t) = seasonal component, which models the periodic effects (e.g weekly, yearly)

h(t) = holiday effect, which models the impact of holidays or special events

ϵ_t = error term, which accounts for any noise that is not explained by the model

(Taylor, Sean J.,Letham, Benjamin, 2018)

3.3 Data Preparation

Data preprocessing is a crucial step in preparing raw data for analysis and modeling. It involves several tasks, including data cleaning, where missing values are handled through imputation or removal, and outliers are identified and treated to ensure they don't skew the results. The data may also need to be transformed through normalization or scaling to ensure that all features contribute equally to the model. In cases of time series data, preprocessing includes differencing to achieve stationarity, as well as creating lagged features to capture temporal dependencies. Additionally, categorical variables might be encoded into numerical formats, and irrelevant or redundant features are often removed to reduce dimensionality and improve model performance. Performance metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) are then used to evaluate the model's accuracy and effectiveness after preprocessing, ensuring that the data is in a suitable format for accurate and efficient modelling.

3.3.1 Data Collection

This dataset provides a detailed hourly record of electricity consumption and production in Romania, starting from 2019. It contains 46,011 rows and 10 columns, capturing both the demand (consumption) and supply (production) of electricity, along with contributions from various energy sources such as nuclear, wind, hydroelectric, oil and gas, coal, solar, and biomass. The dataset's timestamped entries allow for the analysis of how electricity usage and generation fluctuate over time, highlighting patterns such as peak consumption periods, the variability of renewable energy sources, and the reliance on traditional energy sources. This data is valuable for exploring trends in energy production and consumption and understanding the dynamics of the energy market.

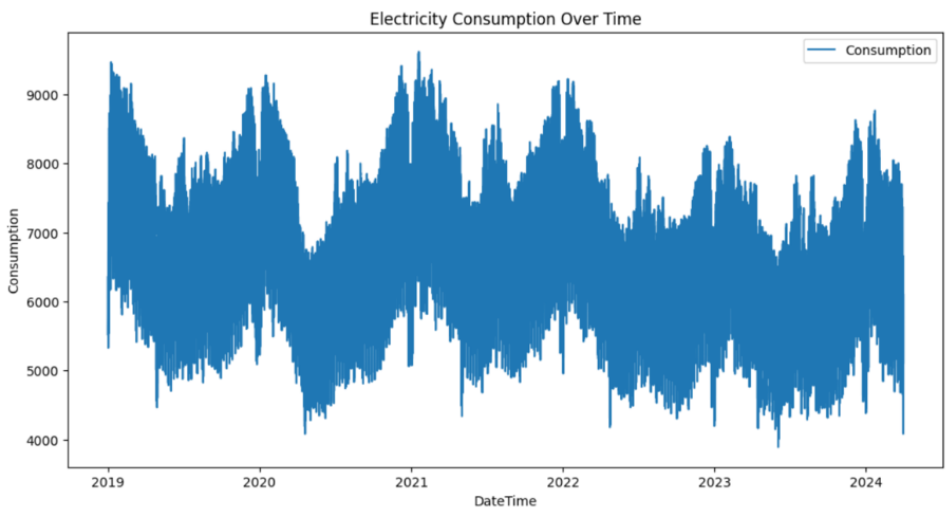

Figure 2 5 Years and 4 months of Electricity Consumption Data in Hourly Basis

```
                     Consumption
DateTime
2019-01-01 00:00:00       6352.0
2019-01-01 01:00:00       6116.0
2019-01-01 02:00:00       5873.0
2019-01-01 03:00:00       5682.0
2019-01-01 04:00:00       5557.0
...                          ...
2024-03-31 19:00:00       5618.0
2024-03-31 20:00:00       6107.0
2024-03-31 21:00:00       5991.0
2024-03-31 22:00:00       5527.0
2024-03-31 23:00:00       5111.0
```

Figure 3 First and last rows of the Dataset

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Data preprocessing is a crucial step in preparing raw data for analysis and modeling. It involves several tasks, including data cleaning, where missing values are handled through imputation or removal, and outliers are identified and treated to ensure they don't skew the results. The data may also need to be transformed through normalization or scaling to ensure that all features contribute equally to the model. In cases of time series data, preprocessing includes differencing to achieve stationarity, as well as creating lagged features to capture temporal dependencies. Additionally, categorical variables might be encoded into numerical formats, and irrelevant or redundant features are often removed to reduce dimensionality and improve model performance. Performance metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) are then used to evaluate the model's accuracy and effectiveness after preprocessing, ensuring that the data is in a suitable format for accurate and efficient modeling.

3.3.2.1 Stationary Test

The Augmented Dickey-Fuller (ADF) test is a key statistical tool used in time series analysis to assess whether a given time series is stationary, which means its statistical properties, such as mean, variance, and autocorrelation, remain consistent over time. Stationarity is a critical assumption for many forecasting models, including ARIMA, as non-stationary data can lead to unreliable predictions. The ADF test works by testing the null hypothesis that the time series contains a unit root, indicating non-stationarity. The test does this by considering an autoregressive model where the time series is regressed on its lagged values, including a constant, a trend term, and lagged differences to account for serial correlation. The ADF test produces a test statistic that is compared against critical values at various significance levels (e.g., 1%, 5%, 10%). If the test statistic is more negative than the

critical value, or if the p-value is below a certain threshold (commonly 0.05), the null hypothesis of a unit root is rejected, suggesting that the time series is stationary. On the other hand, if the null hypothesis cannot be rejected, it implies that the series is non-stationary, and differencing or other transformations may be required to stabilize it before applying forecasting models. The ADF test is therefore a crucial step in the preprocessing of time series data, ensuring that the data is in the proper form for accurate modeling and prediction.

```
ADF Test for Consumption:
ADF Statistic: -14.089104528689031
p-value: 2.741738581138729e-26
Critical Values:
   1%: -3.430492308946249
   5%: -2.8616028975095658
   10%: -2.56680347828538
Consumption is stationary
```

Figure 4 Result of Augmented Dicky Fuller Test

The results of the Augmented Dickey-Fuller (ADF) test strongly indicate that the time series is stationary. With an ADF test statistic of -13.886, which is significantly lower than the critical values at the 1%, 5%, and 10% significance levels, there is clear evidence to reject the null hypothesis of a unit root. The p-value is extraordinarily low, at 6.088e-26, further confirming the rejection of non-stationarity. The test was conducted using 56 lags and 45,945 observations, ensuring robust analysis. These results suggest that the time series does not exhibit a unit root and is stable over time, making it well-suited for modeling without requiring additional transformations like differencing.

3.3.2.2 Checking Missing Values

```
Missing values in each column:
 DateTime         0
Consumption      0
Production       0
Nuclear          0
Wind             0
Hydroelectric    0
Oil and Gas      0
Coal             0
Solar            0
Biomass          0
dtype: int64
```

Figure 5 Result of Checking Missing Values in the Dataset

Checking missing values over the dataset shows that there are no missing values across any columns in the dataset. Specifically, the `DateTime`, `Consumption`, `Production`, `Nuclear`, `Wind`, `Hydroelectric`, `Oil and Gas`, `Coal`, `Solar`, and `Biomass` columns all have a count of zero missing entries. This completeness ensures that the dataset is fully intact and ready for analysis or modeling without the need for imputation or data cleaning related to missing values.

3.3.2.3 Outliers Removal

3.3.2.3.1 Z-score Calculation

```
# Z-score Calculation
def z_score(series):
    mean = series.mean()
    std_dev = series.std()
    z_scores = (series - mean) / std_dev
    return z_scores

# Calculate Z-scores for the 'y' column
df1['zscore'] = z_score(df1['y'])

# Define a custom threshold for outliers
custom_threshold = 2.5

# Identify outliers
outliers = df1[np.abs(df1['zscore']) > custom_threshold]

if not outliers.empty:
    print("Outliers detected:")
    print(outliers[['ds', 'y', 'zscore']])
else:
    print("No outliers detected.")
```

Figure 6 Searching Outliers in Prophet Model

The Z-score, or standard score, measures how many standard deviations a data point is from the mean of a dataset. A Z-score of 0 indicates that the data point is exactly at the mean, while positive Z-scores indicate values above the mean and negative Z-scores represent values below the mean. Z-scores are commonly used for outlier detection, standardizing data for comparison, and understanding probabilities in a normal distribution. Values beyond ±3 are often considered outliers. Effective for identifying outliers in large datasets with sufficient data points. But it is less effective in small datasets or non-normally distributed data; alternative methods like Modified Z-scores may be more robust in such cases. (Bishop, 2006)

$$Z = \frac{x - \mu}{\sigma}$$

where:

Z = standard score

x = observed value

μ = mean of the sample

σ = standard deviation of the sample

3.3.2.3.2 Winsorization

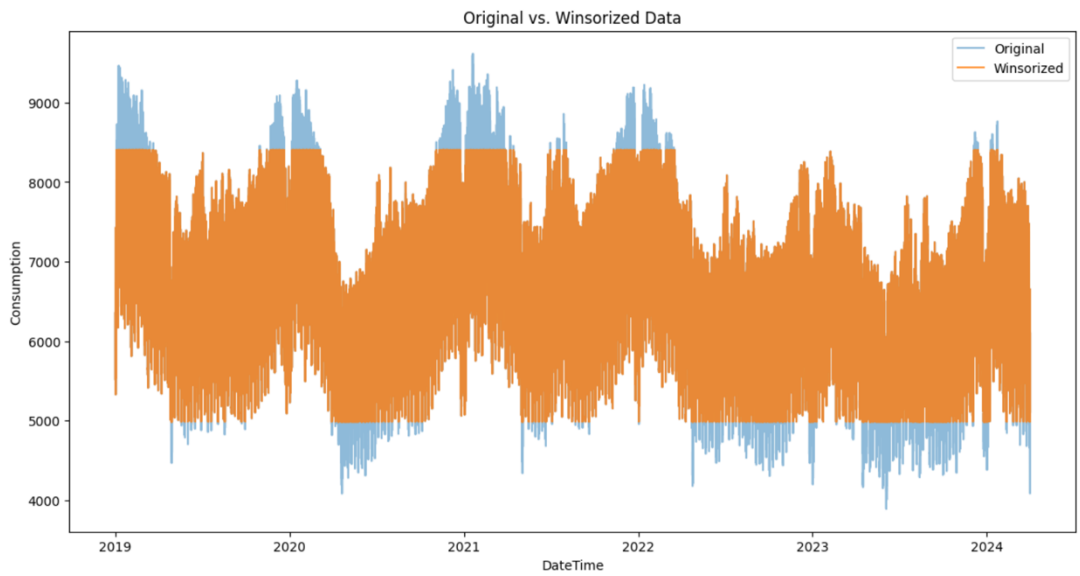

Figure 7 Comparison between Original and Winsorized Data

Winsorization is applied to limit the influence of extreme outliers in the data, particularly in the target variable, “Consumption.” In time series forecasting, extreme values or outliers can skew the model’s predictions, leading to inaccuracies in performance metrics like MSE and MAE. By winsorizing the data, the extreme values are replaced with less extreme values, which makes the distribution more robust to outliers without completely removing them. (Boris Iglewicz , David Caster Hoaglin, 1993)

The graph showcases original and winsorized electricity consumption data from 2019 to 2024, highlighting how winsorization is employed to smooth extremes for enhanced stability in analysis. Both datasets exhibit distinct seasonal variations, with consumption peaks generally occurring in colder months, indicative of increased energy usage. The winsorized data, depicted in orange, effectively moderates the extremes seen in the original blue data by capping the peaks and elevating the troughs, which aids in minimizing the influence of outliers. This technique bolsters model accuracy by ensuring that the data conforms more closely to central trends and lessens the impact of atypical spikes that might skew analysis. Although the data has been adjusted, the overall consumption trend and seasonal patterns remain consistent across both sets, illustrating that winsorization preserves the fundamental behavior of the data while making it more robust for analytical purposes.

Ultimately, this helps improve the reliability and stability of the SARIMAX model, leading to more accurate and meaningful predictions, particularly in scenarios where outliers could distort the forecast.

3.4 Performance Metrics

Performance metrics are essential tools for evaluating the accuracy and effectiveness of predictive models, particularly in regression analysis. In this report, we will discuss three commonly used performance metrics: Mean Squared Error (MSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). Each of these metrics provides a different perspective on the errors made by a predictive model.

1. **Mean Squared Error (MSE)**

Mean Squared Error (MSE) is the average of the squared differences between the actual and predicted values. It is calculated as:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

where:

n = number of observations

y_i = actual value

$\hat{y}_i$ = predicted value

2. **Mean Absolute Error (MAE)**

Mean Absolute Error (MAE) is the average of the absolute differences between the actual and predicted values. It is calculated as:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

where:

n = number of observations

y_i = actual value

$\hat{y}_i$ = predicted value

3. **Mean Absolute Percentage Error (MAPE)**

Mean Absolute Percentage Error (MAPE) is the average of the absolute percentage differences between the actual and predicted values. It is calculated as:

$$MAPE = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|$$

where:

n = number of observations

y_i = actual value

$\hat{y}_i$ = predicted value

The choice of performance metric depends on the specific needs of the problem at hand. MSE is favored in situations where large errors must be penalized more, MAE is preferred for its simplicity and interpretability, and MAPE is suitable for percentage-based error measurement but must be used with caution. (Bishop, 2006)

Chapter 4

Performance Analysis

4.1 Train & Test Split

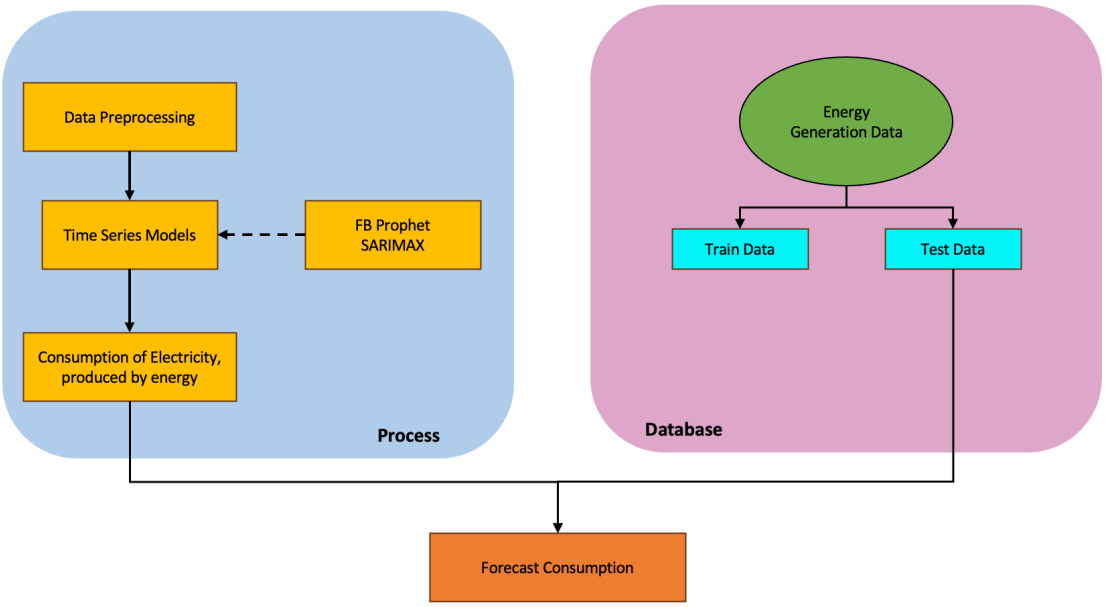

Figure 8 Flowchart of Energy Consumption Prediction

Fig 8 illustrates a renewable energy forecasting process. On the right, electricity generation data is stored in a database, which is split into train and test datasets. These datasets feed into the process (on the left) where data preprocessing occurs first, ensuring the data is cleaned and ready for model training. Time series models, such as Facebook Prophet (FB Prophet) and SARIMAX, are then applied to forecast future electricity consumption based on historical trends. The models are trained on the pre-processed data, with predictions iteratively refined. The output of this pipeline is a future renewable electricity consumption prediction, which can be utilized for planning and decision-making in energy management systems.

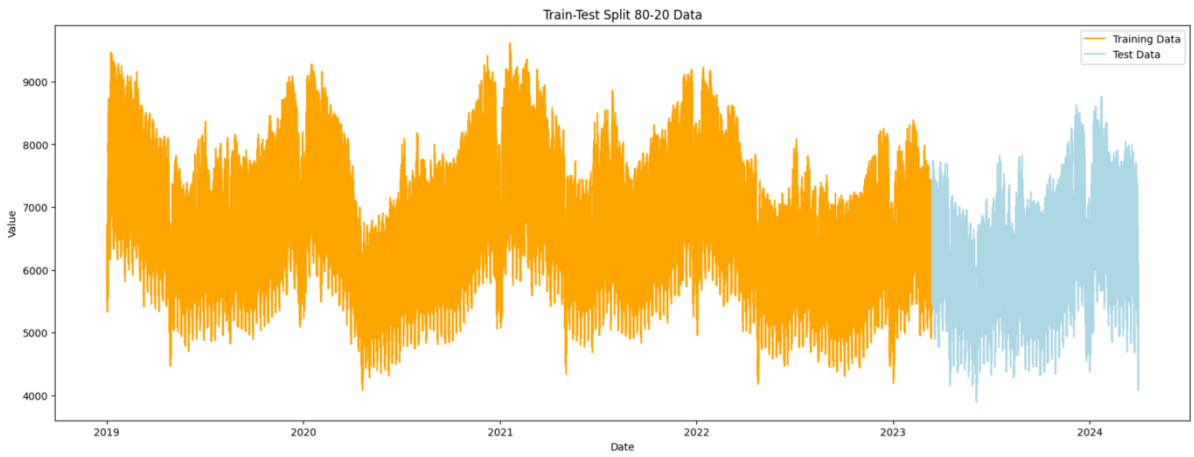

Figure 9 Train-Test Split

In figure 9, the graph likely depicts a time series of consumption of electricity produced by energy sources. It shows a clear seasonal pattern, likely influenced by factors like daylight hours, wind patterns, or water availability. There appears to be a slight upward trend, suggesting increasing reliance on energy.

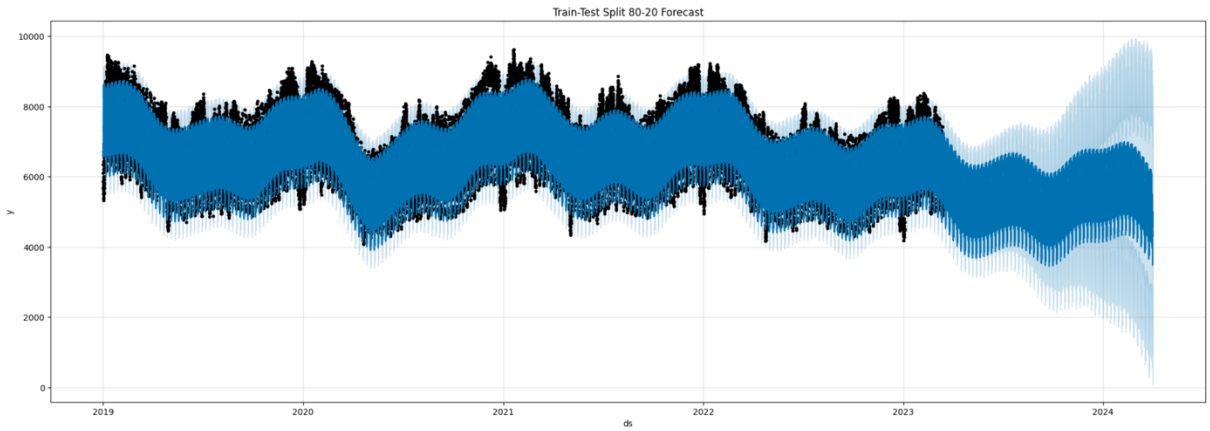

Figure 10 Forecast using FBProphet, with Train-Test Split of 80-20

In figure 10, the graph presents a forecast of electricity consumption produced by renewable energy sources using Facebook Prophet. The black dots represent the actual data, while the blue line shows the predicted values. The shaded area indicates the uncertainty range (prediction interval). The model appears to capture the data's trend and seasonality reasonably well, with some discrepancies between predicted and actual values. The prediction interval is relatively wide, suggesting uncertainty in the forecast.

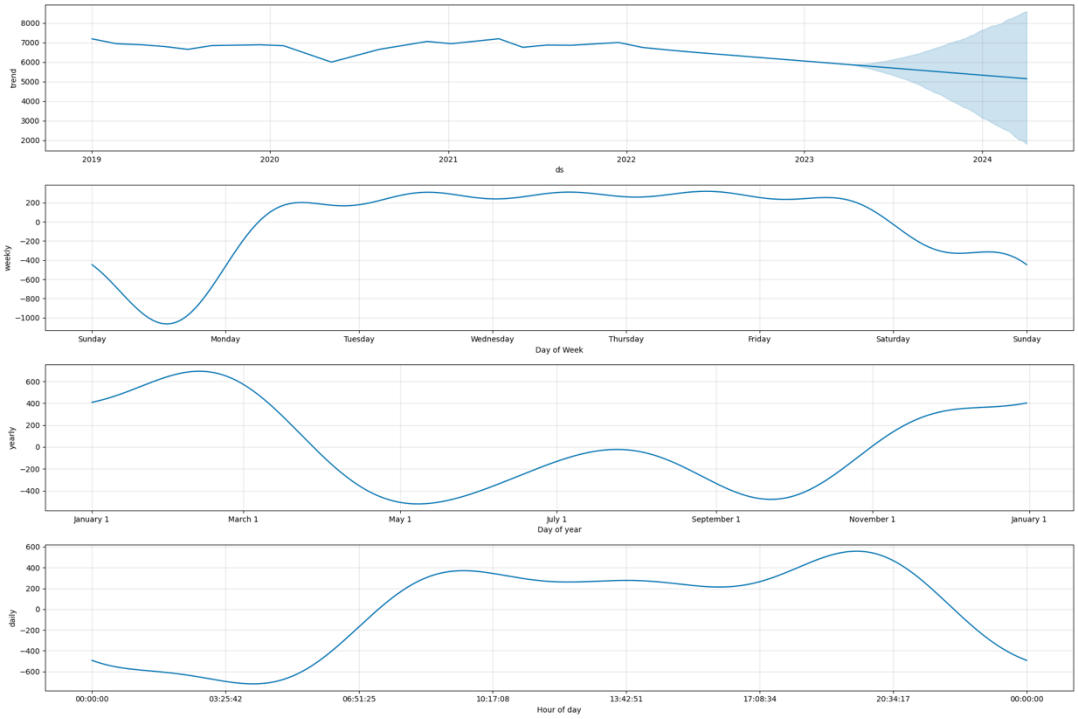

Figure 11 Trend, Weekly, Yearly, and Daily Components Plots

The graph in above figure shows a Facebook Prophet forecast with four component plots. The top plot shows the overall trend, indicating a slight downward trend over time. The second plot shows the weekly seasonality, with a peak on Sundays and a trough on Mondays. The third plot shows the yearly seasonality, with a peak in the summer and a trough in the winter. The fourth plot shows the daily seasonality, with a peak in the early morning and a trough in the late afternoon.

Performance Metrics from Train-Test Splits			
Split	MSE	MAE	MAPE
80-20	6.15E+05	660.79	10.67%
70-30	5.08E+05	566.53	9.15%
60-40	5.58E+06	2108.48	34.67%
50-50	2.30E+07	4242.31	69.51%
40-60	4.42E+06	1792.7	29.45%

Based on the evaluation metrics, the 70-30 train-test split provides the best performance for the model. It achieves the lowest Mean Squared Error (MSE) of 5.08e+05, Mean Absolute Error (MAE) of 566.53, and Mean Absolute Percentage Error (MAPE) of 9.15%, indicating more accurate predictions compared to other splits. While the 80-20 split also performs well, the 70-30 split slightly outperforms it in all key metrics. The larger splits, such as 60-40, 50-50, and 40-60, show significantly higher error values, particularly with large increases in MSE, MAE, and MAPE, suggesting that they are less effective for this model. Therefore, the 70-30 split is the optimal choice for accurate forecasting.

4.2 Cross-Validation

Cross-validation is an essential technique in machine learning for evaluating a model's performance and its ability to generalize to new, unseen data. It involves splitting the dataset into multiple subsets, training the model on one part, and testing it on another. This process is repeated several times to provide a more robust evaluation than a single train-test split. In k-fold cross-validation, the dataset is divided into k equal-sized folds, with the model being trained on k-1 folds and tested on the remaining fold. This rotation helps average the results, offering a reliable measure of model accuracy and reducing the bias of a single train-test split. Cross-validation is particularly beneficial for small datasets, as it makes full use of available data and helps mitigate overfitting—where the model performs well on training data but poorly on unseen data. It provides a more accurate estimate of real-world performance and facilitates the comparison of different models or parameter settings.

For time series data, traditional cross-validation methods like k-fold are not suitable because they disrupt the temporal order of the observations, which is crucial for forecasting. Instead, time series cross-validation techniques, such as rolling window cross-validation or walk-forward validation, are used. These methods respect the sequential nature of the data by ensuring that past data is used to predict future outcomes. Rolling window cross-validation involves training and testing on a moving window of data, while walk-forward validation updates the training set incrementally as the forecast horizon advances. These approaches better reflect real-world scenarios where historical data is used to forecast future values, ensuring that the model's evaluation remains relevant and realistic. (Bishop, 2006)

4.2.1 Time Series Cross Validation

Time series cross-validation is a method designed to evaluate forecasting models while respecting the sequential nature of time series data. Unlike traditional cross-validation, which randomizes data splits, time series cross-validation maintains the temporal order by sequentially splitting the data into training and testing sets. This approach often uses techniques like rolling windows, where a fixed-size training period moves forward in time, or expanding windows, where the training set grows while the test set remains fixed. By simulating real-world forecasting conditions and avoiding data leakage, time series cross-validation provides a realistic assessment of a model's performance, helping ensure that it accurately predicts future values based on historical data. This method is crucial for applications such as sales forecasting and financial analysis, where preserving the chronological order of data is essential for accurate predictions. (Box, George E. P.; Jenkins, Gwilym M.; Reinsel, Gregory C., 2008)

4.2.1.1 TimeSeriesSplit

TimeSeriesSplit is a specific implementation of time series cross-validation provided by libraries like scikit-learn. It splits data sequentially, maintaining the temporal order. Unlike traditional cross-validation, which shuffles data randomly, TimeSeriesSplit divides the dataset into a series of sequential folds where each training set includes all data up to a certain point in time, and each test set comprises subsequent observations. This method ensures that future values are never used to train the model, thereby avoiding data leakage and providing a realistic assessment of forecasting performance. By expanding the training set as time progresses and evaluating the model on contiguous future periods, TimeSeriesSplit offers a robust measure of how well a model generalizes to unseen data, making it a crucial tool for tasks such as sales forecasting, financial analysis, and weather prediction.

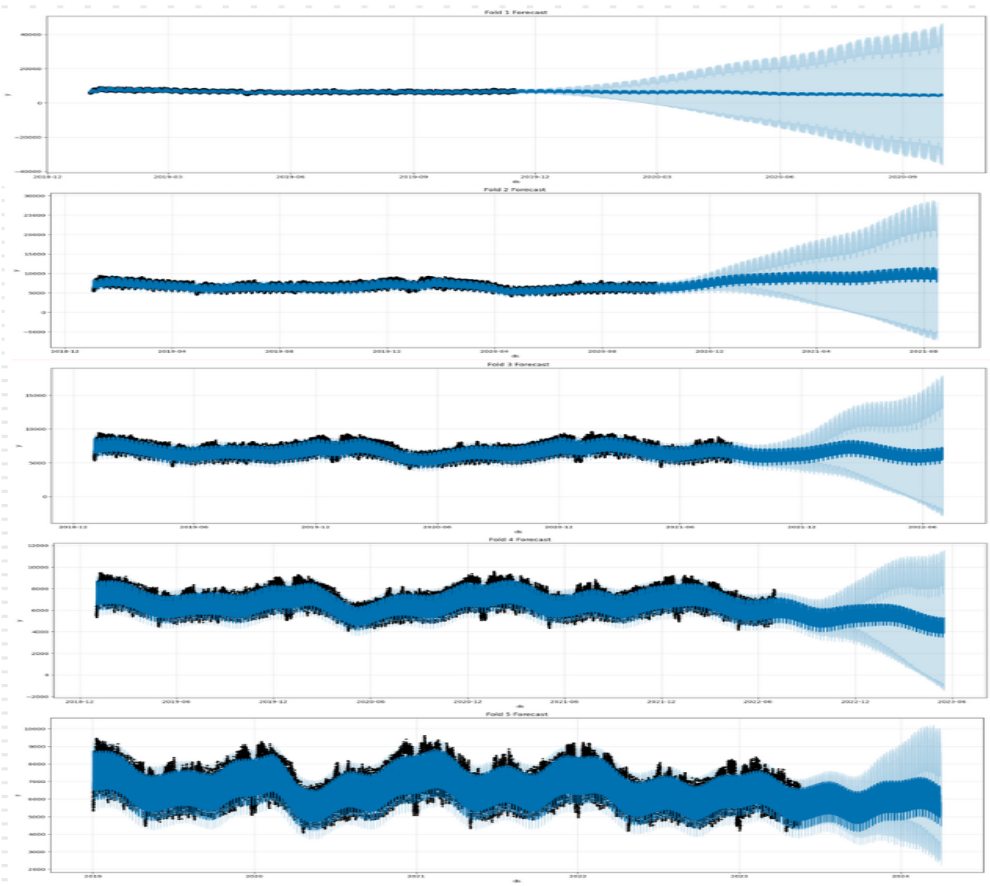

Figure 12 Time Series Cross Validation using 5 Splits (in FB Prophet)

The graphs in figure 12 represents the results of a time series cross-validation for time series forecasting, with each plot showing the predictions and confidence intervals for a different fold. The black lines depict actual data, while the blue shaded areas represent the model's forecasts along with the uncertainty bounds. As seen across the splits, the predictions show increasing uncertainty over time, indicated by the widening of the confidence intervals. The middle splits highlight a clear cyclical or seasonal pattern, suggesting the presence of repeating trends in the data. While some splits exhibit relatively stable trends, others show more variability, indicating the model is capturing different behaviours in various segments of the dataset. This cross-validation approach ensures the model's robustness by testing its performance on different portions of the time series, helping assess its generalization capabilities.

```python
def time_series_cv(df, n_splits=5):
    tscv = TimeSeriesSplit(n_splits=n_splits)
    metrics = []

    for fold, (train_index, val_index) in enumerate(tscv.split(df)):
        print(f"Fold {fold + 1}")
        train_set = df.iloc[train_index]
        val_set = df.iloc[val_index]

        # Initialize and fit the Prophet model
        model = Prophet(seasonality_mode='multiplicative', yearly_seasonality=4, weekly_seasonality=6, daily_seasonality=True)
        model.add_seasonality(name='daily_weekend', period=1, fourier_order=3, condition_name='weekend')
        model.add_seasonality(name='daily_weekday', period=1, fourier_order=3, condition_name='weekday')
        model.fit(train_set)

        # Make predictions
        future = model.make_future_dataframe(periods=len(val_set), freq='H')
        future['weekend'] = future['ds'].apply(lambda x: x.dayofweek >= 5)
        future['weekday'] = ~future['ds'].apply(lambda x: x.dayofweek >= 5)

        forecast = model.predict(future)

        # Evaluate performance
        actual = val_set[['ds', 'y']].set_index('ds')
        predicted = forecast[['ds', 'yhat']].set_index('ds')
        performance = actual.join(predicted).dropna()

        mse = np.mean((performance['y'] - performance['yhat']) ** 2)
        mae = np.mean(np.abs(performance['y'] - performance['yhat']))
        mape = np.mean(np.abs((performance['y'] - performance['yhat']) / performance['y'])) * 100

        metrics.append({
            'fold': fold + 1,
            'mse': mse,
            'mae': mae,
            'mape': mape
        })

        # Plot forecast and components
        fig = model.plot(forecast, figsize=(20, 7))
        plt.title(f'Fold {fold + 1} Forecast')
        plt.show()

        fig2 = model.plot_components(forecast, figsize=(20, 20))
        plt.show()

    return pd.DataFrame(metrics)
```

Figure 13 Code for Time Series Cross Validation using FBProphet

The code in figure 13 executes time series cross-validation on time series data using Prophet model to assess its forecasting performance. It begins by splitting the dataset into training and test subsets for each fold using time series cross-validation. For each fold, the model is trained on the training subset and then used to forecast future values for the length of the test subset. Conditional seasonalities are included in the model, such as weekend and weekday effects. After forecasting, the actual versus predicted values are compared, and performance metrics—Mean Squared Error (MSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE)—are calculated and stored. Finally, the forecasts and their components are plotted for each fold to visualize the model's performance and seasonal patterns.

Performance Metrics from Time Series Cross Validation (Facebook Prophet)			
Fold	MSE	MAE	MAPE
1	1.12E+06	882.35	13.37%
2	4.41E+06	1739.17	25.81%
3	3.39E+05	432.63	6.24%
4	7.34E+05	730.08	11.47%
5	3.14E+05	431.89	6.85%

Based on the performance metrics, Fold 3 is the most favourable among the five folds. It exhibits the lowest Mean Squared Error (MSE) and Mean Absolute Error (MAE), along with a relatively low Mean Absolute Percentage Error (MAPE). This indicates that the forecasts for Fold 3 are the most accurate and consistent, reflecting the model's ability to predict future values with minimal error. In contrast, Fold 2 shows significantly higher errors, making it the least reliable. Therefore, Fold 3 provides the best overall performance and can be considered the most reliable evaluation of the model's forecasting accuracy.

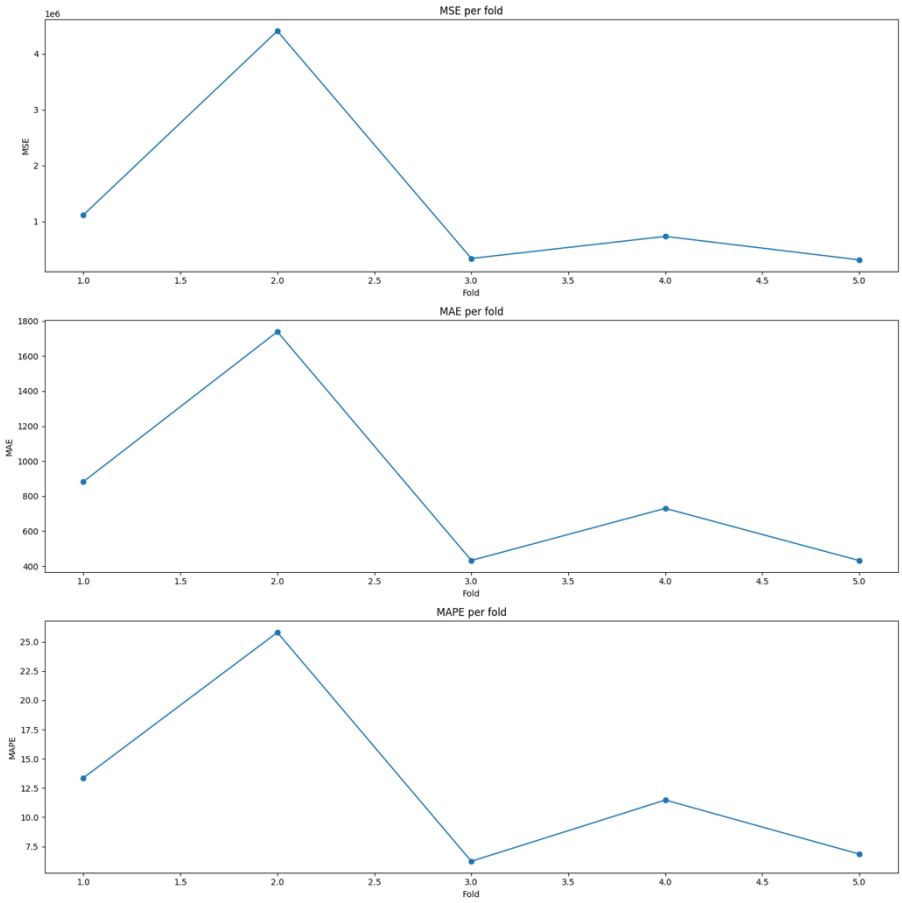

Figure 14 Performance Metrics Plots

Performance Metrics from Time Series Cross Validation (SARIMAX)			
Fold	MSE	MAE	MAPE
1	8.11E+05	727.0	11.66
2	1.06E+06	830.9	11.40
3	6.52E+05	638.8	9.61
4	5.09E+05	546.2	9.20
5	7.22E+05	658.4	10.30

Based on the metrics, Fold 4 performs the best. It has the lowest values in MSE (5.090836e+05), MAE (546.219649), and MAPE (9.204811%). These metrics indicate that Fold 4 has the least squared error, average absolute error, and percentage error among all folds, reflecting superior predictive accuracy. Consequently, Fold 4 is the most effective in terms of error metrics, making it the preferable choice for model evaluation.

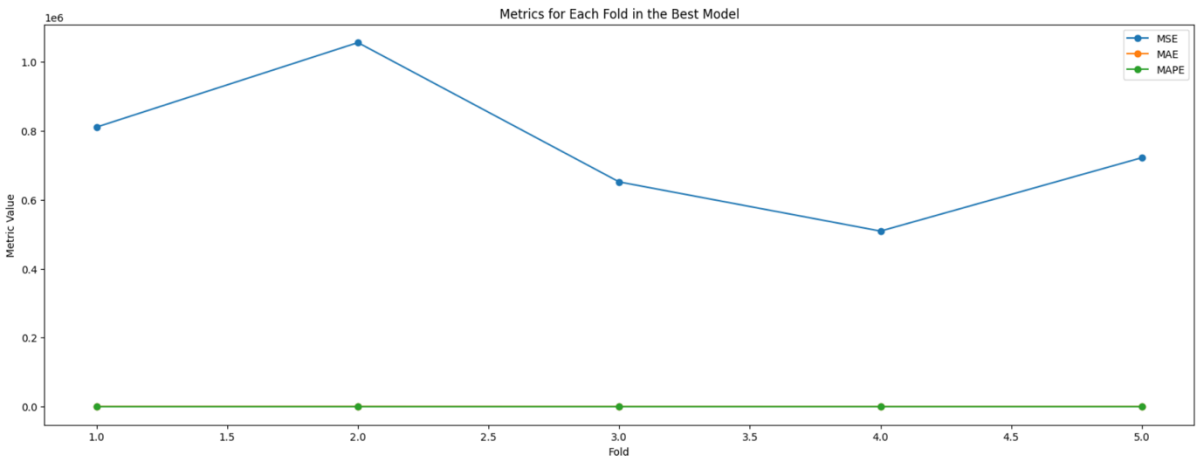

Figure 15 Plotting Metrics for Each Fold in the Best SARIMAX Model

4.3 Parameter Tuning

Parameter tuning, also known as hyperparameter optimization, involves adjusting the settings or parameters of a machine learning model to improve its performance. These parameters, which are not learned from the data but set prior to training, can significantly influence the model's accuracy, generalization, and efficiency. The goal of parameter tuning is to find the optimal combination of these parameters that results in the best performance on validation data. This process often involves techniques such as grid search, random search, and more advanced methods like Bayesian optimization. Effective parameter tuning helps in enhancing the model's ability to learn from data and make accurate predictions. (Géron, 2019)

4.3.1 Tuning Facebook Prophet

In time series modeling with tools like Facebook Prophet, several key parameters are essential for optimizing model performance. Seasonality Parameters include yearly, weekly, and daily seasonality, and can be extended to custom seasonalities such as weekend or weekday effects, helping capture various periodic patterns in the data. Fourier Orders control the flexibility of these seasonal components, with higher orders allowing the model to fit more complex seasonal patterns through additional sine and cosine terms. Changepoint Parameters, particularly changepoint_prior_scale, adjust the model's responsiveness to trend changes, balancing between capturing significant shifts and smoothing out minor fluctuations. The Seasonality Mode offers a choice between 'additive' and 'multiplicative' approaches: 'additive' assumes seasonal effects are constant, while 'multiplicative' assumes they scale with the trend. Tuning these parameters is crucial for accurately capturing underlying data patterns without overfitting. (Rob J Hyndman and George Athanasopoulos, 2021)

The following parameters are used for tuning this Facebook Prophet model:

- seasonality_mode: ['additive', 'multiplicative'],
- yearly_seasonality: [True],
- weekly_seasonality: [True],
- daily_seasonality: [True],
- changepoint_prior_scale: [0.1, 1.0],
- fourier_order: [3, 5, 7]

To evaluate the effects of different parameter settings on the Prophet model, a subset of results from multiple parameter tuning experiments will be presented. These results are selected randomly to offer a representative overview of how varying parameters influence the model's performance. This approach ensures a broad perspective on the model's behavior across different configurations.

For: Changepoint Prior Scale: 1.0
Daily Seasonality: True
Fourier Order: 7
Seasonality Mode: additive
Weekly Seasonality: True
Yearly Seasonality: True

Performance Metrics from Parameter Tuning (Facebook Prophet)			
Fold	MSE	MAE	MAPE
1	7.61E+09	84277.96	1332.4795
2	7.38E+06	2309.66	33.8123
3	6.56E+06	2174.12	33.2473
4	1.89E+05	338.48	5.6263
5	2.40E+05	378.70	6.2622

For: Changepoint Prior Scale: 0.1
Daily Seasonality: True
Fourier Order: 3
Seasonality Mode: additive
Weekly Seasonality: True
Yearly Seasonality: True

Performance Metrics from Parameter Tuning (Facebook Prophet)			
Fold	MSE	MAE	MAPE
1	5.23E+09	69337.9	1101.61
2	5.34E+05	534.0	7.69
3	1.12E+06	920.2	14.24
4	2.17E+05	367.6	6.14
5	2.42E+05	382.4	6.29

For: Changepoint Prior Scale: 1.0

Daily Seasonality: True

Fourier Order: 3

Seasonality Mode: multiplicative

Weekly Seasonality: True

Yearly Seasonality: True

Performance Metrics from Parameter Tuning (Facebook Prophet)			
Fold	MSE	MAE	MAPE
1	1.64E+07	3355.5	52.81
2	2.45E+06	1274.4	18.98
3	6.06E+05	651.6	9.36
4	1.98E+06	1252.4	19.83
5	3.56E+05	477.0	7.52

For: Changepoint Prior Scale: 0.1

Daily Seasonality: True

Fourier Order: 5

Seasonality Mode: additive

Weekly Seasonality: True

Yearly Seasonality: True

Performance Metrics from Parameter Tuning (Facebook Prophet)			
Fold	MSE	MAE	MAPE
1	5.19E+09	69129.9	1098.13
2	4.76E+05	558.7	8.34
3	9.12E+05	814.8	12.62
4	1.89E+05	339.2	5.64
5	2.41E+05	382.8	6.29

For: Changepoint Prior Scale: 0.1
Daily Seasonality: True
Fourier Order: 7
Seasonality Mode: multiplicative
Weekly Seasonality: True
Yearly Seasonality: True

Performance Metrics from Parameter Tuning (Facebook Prophet)			
Fold	MSE	MAE	MAPE
1	9.39E+06	2684.9	42.34
2	5.51E+06	1928.6	28.59
3	3.06E+05	433.8	6.24
4	1.09E+06	911.1	14.35
5	2.95E+05	428.1	6.80

Among the above parameters, Changepoint Prior Scale: 1.0, Daily Seasonality: True, Fourier Order: 7, Seasonality Mode: additive, Weekly Seasonality: True, Yearly Seasonality: True is the best parameter tuning for this Facebook Prophet model.

4.3.2 Tuning SARIMAX Model

4.3.2.1 Randomized Search

In the process of tuning a SARIMAX model, random search is a practical approach for optimizing hyperparameters. Unlike exhaustive methods, random search samples a subset of possible parameter combinations randomly, allowing for a broader exploration of the parameter space without requiring a comprehensive grid search. This method is particularly useful for complex models with many hyperparameters, as it can efficiently identify effective parameter settings while minimizing computational expense. Random search can be implemented to fine-tune parameters such as the seasonal order, non-seasonal order, and exogenous variables in the SARIMAX model. This approach balances exploration and exploitation, making it a valuable tool for improving model performance. (Bovas Abraham, Johannes Ledolter, 2009)

In this parameter tuning, random search involves randomly selecting parameter combinations from predefined ranges for the SARIMAX model. These ranges are set for the non-seasonal parameters p, d , and q , as well as the seasonal parameters P , D , Q , and m , with the seasonal period fixed at 24 to account for daily seasonality in an hourly time series. To assess efficiency, five random combinations are tested, allowing for an exploration of different parameter settings without exhaustive computation.

```
p_range = [0, 1, 2]
d_range = [0, 1]
q_range = [0, 1, 2]
seasonal_p_range = [0, 1]
seasonal_d_range = [0, 1]
seasonal_q_range = [0, 1]
seasonal_period_range = [24]
n_combinations = 5
```

Figure 16 Parameters Tuning SARIMAX model

The following orders are randomly searched by tuning the parameters:

Model Order	Seasonal Order
• (1, 0, 0)	(0, 1, 0, 24)
• (0, 1, 0)	(0, 0, 0, 24)
• (2, 0, 2)	(0, 1, 1, 24)
• (0, 1, 0)	(0, 1, 0, 24)
• (2, 1, 0)	(1, 0, 0, 24)

For: (1, 0, 0) (0, 1, 0, 24)

Performance Metrics from Parameter Tuning (SARIMAX)			
Fold	MSE	MAE	MAPE
1	1.11E+06	841.2	12.18
2	9.34E+05	775.6	10.90
3	4.89E+06	2047.2	32.08
4	1.43E+06	963.8	16.60
5	6.02E+05	603.0	10.00

For: (0, 1, 0) (0, 0, 0, 24)

Performance Metrics from Parameter Tuning (SARIMAX)			
Fold	MSE	MAE	MAPE
1	1.29E+06	947.5	15.64
2	3.70E+06	1640.7	21.74
3	2.54E+06	1349.3	22.33
4	8.25E+05	770.8	12.59
5	1.10E+06	875.7	15.49

For: (2, 0, 2) (0, 1, 1, 24)

Performance Metrics from Parameter Tuning (SARIMAX)			
Fold	MSE	MAE	MAPE
1	8.32E+05	734.8	11.82
2	1.07E+06	838.5	11.53
3	6.40E+05	635.1	9.60
4	6.17E+05	594.8	10.24
5	6.84E+05	639.9	10.08

For: (0, 1, 0) (0, 1, 0, 24)

Performance Metrics from Parameter Tuning (SARIMAX)			
Fold	MSE	MAE	MAPE
1	1.77E+10	115509.0	1857.79
2	3.61E+07	5323.4	77.29
3	1.13E+11	291118.7	4481.88
4	7.73E+08	24270.3	400.12
5	2.51E+08	13809.9	225.17

For: (2, 1, 0) (1, 0, 0, 24)

Performance Metrics from Parameter Tuning (SARIMAX)			
Fold	MSE	MAE	MAPE
1	1.29E+06	4097.4	61.42
2	5.00E+06	1973.9	26.47
3	1.01E+08	9969.1	153.18
4	1.27E+06	923.3	16.55
5	1.25E+06	936.0	16.77

(0,1,0) (0,0,0,24) is the best model among those randomly search models, and fold 4 is the best fold for the model.

Chapter 5
Conclusion

5.1 Conclusion

In this report, we explored two prominent time series forecasting models: FB Prophet and SARIMAX, are applied to the dataset called “Hourly Electricity Consumption in Romania”. Both models are designed to handle time series data with trends and seasonality, yet they differ significantly in terms of their approach and underlying assumptions. FB Prophet, developed by Facebook, is a robust model that excels in handling missing data, outliers, and non-linear trends. It is particularly effective when the dataset exhibits strong seasonality and multiple trend components. Its intuitive interface, combined with flexibility in modeling holidays and special events, makes it a powerful tool for practical forecasting applications. SARIMAX (Seasonal AutoRegressive Integrated Moving Average with eXogenous factors), on the other hand, builds upon traditional statistical modeling techniques. It leverages past values (AR), errors (MA), seasonal patterns (S), and external variables (X) to provide a well-rounded forecast. While SARIMAX offers more control over parameter selection and model tuning, it can be less user-friendly, requiring deeper statistical expertise. Throughout the analysis, both models provided valuable insights into the future trends of the dataset. However, the comparative analysis revealed that each model has its strengths depending on the complexity of the data. For datasets with intricate seasonality and irregular trends, FB Prophet demonstrated superior performance due to its adaptability. SARIMAX, with its classical approach, performed well for datasets with more stable and predictable patterns.

In conclusion, the choice between FB Prophet and SARIMAX depends on the nature of the data and the specific forecasting requirements. For users seeking a more automated and flexible solution, FB Prophet is recommended, while SARIMAX is ideal for those requiring greater customization and control over the forecasting process. Further improvements could be explored by hybridizing these models or integrating additional external variables to enhance prediction accuracy.

4.2 Expected Outcomes

The expected outcomes of this project focus on accurately forecasting energy consumption produced by renewable energy sources using SARIMAX and Prophet models. These models will enhance decision-making for managing the grid and allocating resources, optimizing the use of renewable energy such as solar and wind. Accurate predictions will improve grid stability, reduce energy waste, and lower operational costs for both energy producers and consumers. Additionally, the project will support sustainability efforts by minimizing carbon emissions and helping policymakers develop more effective strategies for integrating energy into the grid, fostering a more resilient and environmentally friendly energy infrastructure.

4.3 Potential Applications

The potential applications of accurate forecasting for energy consumption are vast and transformative, spanning across energy management, grid optimization, and various industries.

1. Energy Grid Management: Grid operators can use these forecasts to optimize the integration of renewable energy sources like solar and wind, ensuring a more balanced energy supply. By predicting fluctuations in renewable energy output, operators can adjust other energy sources to stabilize the grid, prevent outages, and reduce reliance on non-renewable sources. This leads to more efficient energy distribution and minimizes the risk of energy shortages during peak demand periods.

2. Demand Response: Forecasting helps energy companies adjust pricing strategies in real-time based on predicted availability from various energy sources. For example, when solar and wind production is high, prices can be adjusted to encourage consumption during these periods, promoting the use of green energy and optimizing the utilization of renewable resources.

3. Energy Storage Optimization: Forecasts guide decisions on when to store energy from various sources. For instance, excess solar or wind energy can be stored in batteries for use during periods of low production, ensuring a steady supply. This also helps optimize the use of storage solutions and reduces waste from energy production.

4. Agriculture: Farmers can utilize forecasts to align energy-intensive processes, such as irrigation and greenhouse management, with periods of high renewable energy output. This reduces energy costs and promotes sustainable practices by leveraging cleaner energy sources like solar and wind.

5. Electric Vehicle (EV) Charging Infrastructure: Forecasting renewable energy availability helps in planning EV charging station operations. By scheduling charging during peak renewable energy periods, operators can maximize the use of clean energy, minimize grid stress, and support the development of a sustainable transportation infrastructure.

6. Urban Planning and Smart Cities: Urban planners can incorporate energy consumption forecasts into the design of sustainable infrastructure. This includes solar-powered street lighting, renewable energy-powered public transportation, and efficient energy distribution networks. Such planning promotes energy independence and resilience in cities, relying more on clean energy sources.

7. Policy Development: Accurate forecasts provide valuable data for policymakers to develop effective energy policies. Insights into renewable energy consumption trends guide decisions on subsidies, infrastructure investments, and long-term sustainability goals. This supports the adoption of renewable energy and helps in reducing overall carbon emissions.

Overall, the project's predictive capabilities will support a wide range of applications that optimize energy usage, reduce operational costs, and contribute to environmental sustainability across sectors.

https://www.kaggle.com/code/jaysithu/facebook-prophet

https://www.kaggle.com/code/jaysithu/sarimax

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